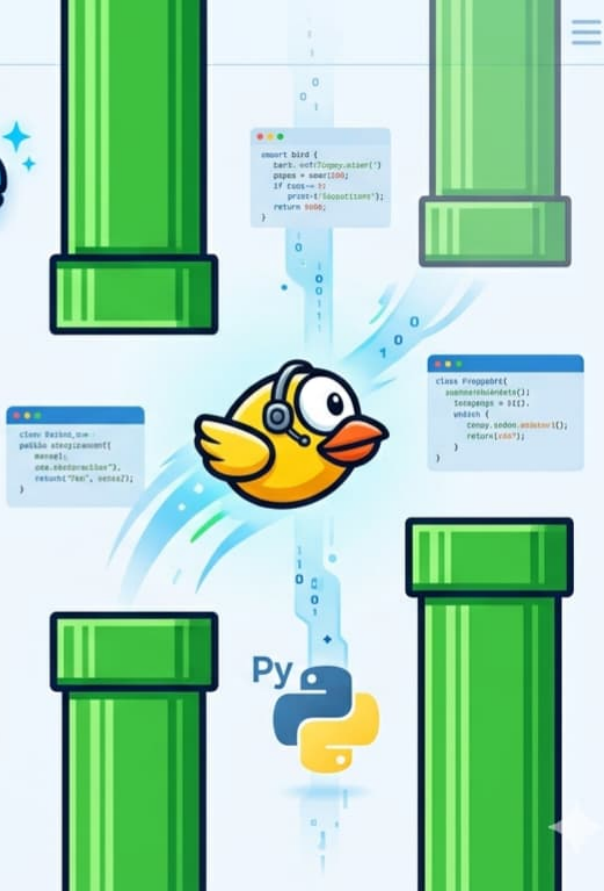
Flappy Bird Game Using Python

- Simple yet engaging arcade style game.
- Control a bird that must pass through pipes without colliding.
- Objective is to achieve the highest possible score by successfully navigating gaps.
- Project demonstrates Python usage with basic programming and graphical libraries.



OBJECTIVE, TOOLS & CONCEPT

OBJECTIVE

- Design and develop a Flappy Bird game using Python
- Focus on implementing game logic, user interaction, collision detection, and animation
- Understand event handling and real-time rendering

TOOLS & TECHNOLOGIES

- Python programming language
- Pygame library handles graphics, animations, and input
- Functionalities for rendering images, keyboard input, and game loops

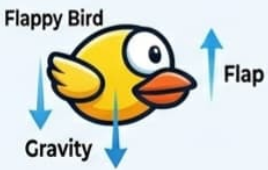
GAME CONCEPT

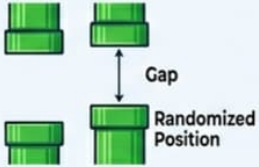
- Player controls a bird that falls due to gravity
- Press a key to make the bird flap upward
- Pipes appear from the right, moving left with random gaps
- Avoid colliding with pipes or the ground to prevent game over

SYSTEM DESIGN: COMPONENTS & GAME LOOP

GAME COMPONENTS

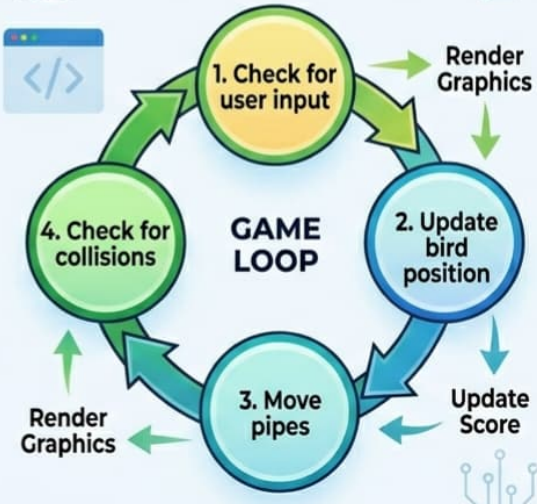
- **Bird:** The player-controlled character
- **Pipes:** Obstacles that appear at regular intervals





- **Score System:** Increases when the bird passes through pairs of pipes

GAME LOOP



WORKING PRINCIPLE & ALGORITHM FLOW

WORKING PRINCIPLE

1. Initialization
- Set up - assets and variables on objects

2. User Input
- Press the larget input ky target

SPACE

3. Movement & Physics

4. Collision Detection
- gravity pulls down, key makes flap up

4. Collision Detection

5. Rendering
- Smooth animation, birds, pipes, and score

ALGORITHM FLOW

Start the game
↓
Initialize variables and load assets
↓
Enter game loop
↓
Check for user input
↓
Update bird position
↓
Move pipes
↓
Check for collisions
↓
Update score if passed
↓
Render all elements
↓
Repeat loop until collision
↓
End game and display final score

FEATURES, ADVANTAGES & LIMITATIONS

FEATURES

- Simple and intuitive gameplay
- Real-time animation
- Score tracking system
- Increasing difficulty
- Interactive user control

ADVANTAGES

- Helps in learning game development basics
- Improves understanding of event-driven programming
- Enhances problem-solving and logical thinking skills

LIMITATIONS

- Limited graphics
- Performance depends on system capability
- Basic gameplay without advanced features like levels or sound effects

FUTURE ENHANCEMENTS & CONCLUSION

FUTURE ENHANCEMENTS

- Adding sound effects and background music
- Introducing different difficulty levels
- Adding start menu and pause functionality
- Improving graphics and animations
- Implementing high score saving system

HIGH SCORE

CONCLUSION

- Great beginner project to learn game development basics
- Combines logic, and interactivity
- Foundational for building complex games
- Provides an enjoyable user experience

COMPLEX GAMES